

Whats Cookin

Feature Testing and Stress Testing Report

CS160A Course Project



Part 1: Functional Feature Testing

This section documents functional testing performed against the current implementation, covering authentication, pantry management, recipe discovery, and the external recipe API integration.

Feature Tested	Testing Scenario	Expected Result	Actual Result	Status
Signup	Submit valid email, password (6+ chars), and matching confirmation	Account is created and user is redirected to the Home page	Account created and redirect to Home occurred as expected	Pass
Signup, mismatched passwords	Submit a password and confirmation that do not match	Form displays a validation message and does not submit	Form displayed a mismatch message and blocked submission	Pass
Signup, invalid credentials	Submit a password under 6 characters or a malformed email	Form displays a descriptive error and account is not created	Descriptive error shown; account creation blocked	Pass
Login	Submit a valid, previously registered email and password	User is authenticated and redirected to the Home page	Login succeeded and redirect occurred as expected	Pass
Logout	Select the logout control while signed in	Session ends and protected pages become inaccessible	Session ended; Pantry/Recipe Details became inaccessible	Pass
Protected routes	Attempt to open /pantry or /recipes/:id while signed out	User is redirected away from the protected page	Redirect occurred; protected pages were not reachable	Pass
Add pantry item	Add an ingredient with a name, quantity, unit, and expiry date	Item appears in the pantry list immediately	Item appeared without a manual page refresh	Pass
Edit pantry item	Open an existing item and change its name, quantity, or unit	Updated values are saved and reflected in the item card	Updated values displayed correctly after saving	Pass
Quantity/unit changes	Change only the quantity or only the unit of an existing item	The changed field updates while other fields remain unchanged	Only the targeted field changed; rest of the item was preserved	Pass
Expiry date updates	Edit an item to add, change, or remove its expiry date	The item's expiry status recalculates to match the new date	Status recalculated correctly on save	Pass

Feature Tested	Testing Scenario	Expected Result	Actual Result	Status
Expiry status indicators	Add items dated in the past, within 7 days, and beyond 7 days	Items display as Expired, Expiring Soon, and Fresh respectively	Correct color/status shown for each date range	Pass
Data persistence	Add pantry items, then log out and log back in	Previously added items are still present	Items persisted across the session as expected	Pass
Homepage recipes	Load the Home page as a signed-in user with a non-empty pantry	Recommended, Random, and Use Soon sections populate with recipes	All three sections populated from the recipe service	Pass
Recommended recipes	Compare Recommended Recipes against the current pantry contents	Only recipes with zero missing ingredients are listed	Every listed recipe was fully cookable from the pantry	Pass
Search functionality	Search by a recipe title and by a single ingredient	Matching recipes are returned regardless of pantry cookability	Relevant recipes returned for both title and ingredient searches	Pass
Recipe filtering	Apply a Cuisine filter, a Time of Day filter, and both together	Recipe sections narrow to only matching recipes	Sections narrowed correctly; combined filters applied as AND	Pass
Recipe details page	Open a recipe card from any Home page section	Full recipe view loads with image, ingredients, and instructions	Recipe details loaded correctly with all expected fields	Pass
Missing ingredient calculation	Open a recipe with a pantry missing one or more ingredients	Missing ingredients are flagged individually and counted accurately	Flags and counts matched the pantry contents	Pass
Substitute feature	Select "Substitute" on a missing ingredient with a pantry-available substitute	A substitute list appears and selecting one updates the displayed ingredient	Substitute applied and "Substituted for X" note appeared	Pass
Spoonacular integration	Load recipe sections and confirm data originates from the external API	Recipes reflect live external data, not hardcoded content	Confirmed recipe data was sourced from the API	Pass
API caching	Reload the Home page shortly after an initial load	Previously fetched recipe data loads without a new API request	Cached data was reused within the caching window	Pass
Reduced repeated requests	Perform the same search twice in quick succession	The second identical request is served from cache/deduplication rather than firing a new call	Second request was served without an additional API call	Pass

Part 2: Stress Testing

This section documents stress and edge-case testing intended to confirm the application degrades gracefully rather than failing outright under unusual conditions.

Empty pantry

Expected Behavior: Recipe sections rely on isRecipeCookable/missing-ingredient logic with zero pantry items

Observed Behavior: Recommended Recipes shows no cookable recipes (or an empty state); Random Recipes and search still function normally

Result: Behaved as expected, Recommended Recipes reflected an empty pantry without crashing other sections

Large pantry

Expected Behavior: A pantry with a large number of saved items (dozens of entries)

Observed Behavior: Pantry list renders all items and matching logic continues to run correctly against the larger set

Result: List rendered fully; recipe matching continued to work correctly

Expired ingredients

Expected Behavior: Multiple pantry items with expiry dates already in the past

Observed Behavior: Each is marked Expired and still counted as present for ingredient-matching purposes

Result: Expired items displayed correctly and still counted toward availability

Invalid ingredient names

Expected Behavior: Attempt to submit a pantry item with an empty or unrecognized name

Observed Behavior: Form validation blocks submission with a clear error

Result: Validation blocked the submission as expected

Duplicate pantry items

Expected Behavior: Add the same ingredient name to the pantry more than once

Observed Behavior: Both entries are stored as separate items rather than causing an error

Result: Duplicate entries were stored independently without errors

Empty search queries

Expected Behavior: Clear the search bar entirely

Observed Behavior: The Home page falls back to the default Recommended/Random view rather than an empty result

Result: Default view was restored correctly

Invalid recipe searches

Expected Behavior: Search for a nonsense term unlikely to match any recipe

Observed Behavior: An empty-results state is shown rather than an error or crash

Result: Empty state displayed cleanly

API quota limitations

Expected Behavior: Simulate a failed or rate-limited API response

Observed Behavior: A friendly error message is shown and, where a cached response exists, stale cached data is served instead of failing outright

Result: Friendly error/fallback behavior observed rather than a crash

Refreshing application

Expected Behavior: Reload the browser tab while signed in with pantry data present

Observed Behavior: Session and pantry data are restored via Firebase without requiring re-entry

Result: Session and pantry state restored correctly after reload

Multiple user accounts

Expected Behavior: Log in as two different accounts (in separate sessions) with different pantry contents

Observed Behavior: Each account's pantry and recommendations remain fully isolated from the other

Result: No cross-account data leakage observed

Missing Firebase data

Expected Behavior: Access a recipe details page immediately after login, before the pantry subscription finishes loading

Observed Behavior: A loading state is shown for ingredient availability instead of an incorrect all-missing/all-available flash

Result: Loading state displayed correctly during the brief pantry-load window